

31. Let P and Q be continuous and have continuous first partial derivatives in a simply connected region of the xy -plane. If $\int_A^B P dx + Q dy$ is independent of the path, show that $\oint_C P dx + Q dy = 0$ on every piecewise-smooth simple closed curve C in the region.
32. Let R be a region bounded by a piecewise-smooth simple closed curve C . Show that the coordinates of the centroid of

the region are given by

$$\bar{x} = \frac{1}{2A} \oint_C x^2 dy, \quad \bar{y} = -\frac{1}{2A} \oint_C y^2 dx.$$

33. Find the work done by the force $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$ acting along the cardioid $r = 1 + \cos \theta$.

9.13 Surface Integrals

Introduction In the xy -plane, the length of an arc of the graph of $y = f(x)$ from $x = a$ to $x = b$ is given by the definite integral

$$s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx. \quad (1)$$

The problem in three dimensions, which is the counterpart of the arc length problem, is to find the area $A(s)$ of that portion of the surface S given by a function $z = f(x, y)$ having continuous first partial derivatives on a closed region R in the xy -plane. Such a surface is said to be smooth.

Surface Area Suppose, as shown in **FIGURE 9.13.1(a)**, that an inner partition P of R is formed using lines parallel to the x - and y -axes. P then consists of n rectangular elements R_k of area $\Delta A_k = \Delta x_k \Delta y_k$ that lie entirely within R . Let $(x_k, y_k, 0)$ denote any point in an R_k . As we see in **Figure 9.13.1(a)**, by projecting the sides of R_k upward, we determine two quantities: a portion or patch S_k of the surface and a portion T_k of a tangent plane at $(x_k, y_k, f(x_k, y_k))$. It seems reasonable to assume that when R_k is small, the area ΔT_k of T_k is approximately the same as the area ΔS_k of S_k .

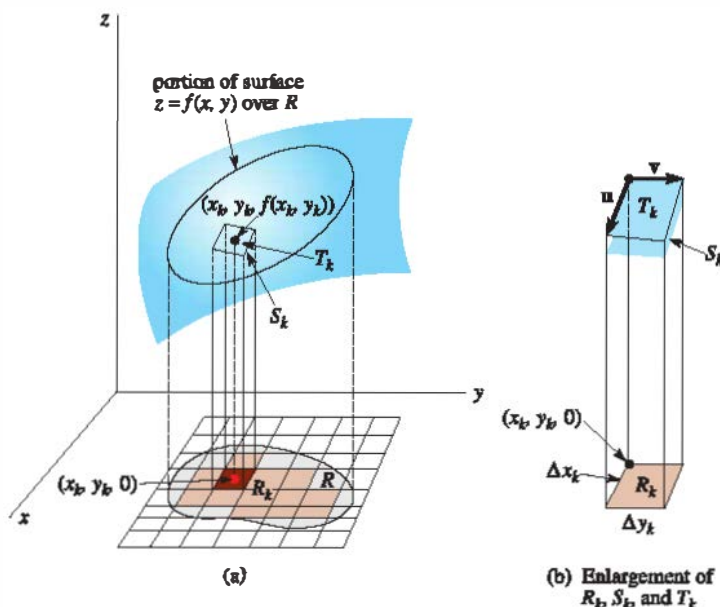


FIGURE 9.13.1 What is the area of the surface above R ?

To find the area of T_k let us choose $(x_k, y_k, 0)$ at a corner of R_k as shown in **Figure 9.13.1(b)**. The indicated vectors $\mathbf{u}$ and $\mathbf{v}$, which form two sides of T_k , are given by

$$\mathbf{u} = \Delta x_k \mathbf{i} + f_x(x_k, y_k) \Delta x_k \mathbf{k}$$

$$\mathbf{v} = \Delta y_k \mathbf{j} + f_y(x_k, y_k) \Delta y_k \mathbf{k},$$

where $f_x(x_k, y_k)$ and $f_y(x_k, y_k)$ are the slopes of the lines containing $\mathbf{u}$ and $\mathbf{v}$, respectively. Now from (16) of Section 7.4 we know that $\Delta T_k = \|\mathbf{u} \times \mathbf{v}\|$ where

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \Delta x_k & 0 & f_x(x_k, y_k)\Delta x_k \\ 0 & \Delta y_k & f_y(x_k, y_k)\Delta y_k \end{vmatrix} = [-f_x(x_k, y_k)\mathbf{i} - f_y(x_k, y_k)\mathbf{j} + \mathbf{k}]\Delta x_k \Delta y_k.$$

In other words,

$$\Delta T_k = \sqrt{[f_x(x_k, y_k)]^2 + [f_y(x_k, y_k)]^2 + 1} \Delta x_k \Delta y_k \\ = \sqrt{1 + [f_x(x_k, y_k)]^2 + [f_y(x_k, y_k)]^2} \Delta A_k.$$

Consequently, the area A is approximately

$$\sum_{k=1}^n \sqrt{1 + [f_x(x_k, y_k)]^2 + [f_y(x_k, y_k)]^2} \Delta A_k.$$

Taking the limit of the foregoing sum as $\|P\| \rightarrow 0$ leads us to the next definition.

Definition 9.13.1 Surface Area

Let f be a function for which the first partial derivatives f_x and f_y are continuous on a closed region R . Then the area of the surface over R is given by

$$A(S) = \iint_R \sqrt{1 + [f_x(x, y)]^2 + [f_y(x, y)]^2} dA. \quad (2)$$

One could have almost guessed the form of (2) by naturally extending the one-variable structure of (1) to two variables.

EXAMPLE 1 Surface Area

Find the surface area of that portion of the sphere $x^2 + y^2 + z^2 = a^2$ that is above the xy -plane and within the cylinder $x^2 + y^2 = b^2$, $0 < b < a$.

SOLUTION If we define $z = f(x, y)$ by $f(x, y) = \sqrt{a^2 - x^2 - y^2}$, then

$$f_x(x, y) = \frac{-x}{\sqrt{a^2 - x^2 - y^2}} \quad \text{and} \quad f_y(x, y) = \frac{-y}{\sqrt{a^2 - x^2 - y^2}}$$

and so
$$1 + [f_x(x, y)]^2 + [f_y(x, y)]^2 = \frac{a^2}{a^2 - x^2 - y^2}.$$

Hence, (2) is
$$A(S) = \iint_R \frac{a}{\sqrt{a^2 - x^2 - y^2}} dA,$$

where R is indicated in **FIGURE 9.13.2**. To evaluate this double integral, we change to polar coordinates:

$$\begin{aligned} A(S) &= a \int_0^{2\pi} \int_0^b (a^2 - r^2)^{-1/2} r dr d\theta \\ &= a \int_0^{2\pi} \left[-(a^2 - r^2)^{1/2} \right]_0^b d\theta = a(a - \sqrt{a^2 - b^2}) \int_0^{2\pi} d\theta \\ &= 2\pi a(a - \sqrt{a^2 - b^2}) \text{ square units.} \end{aligned}$$

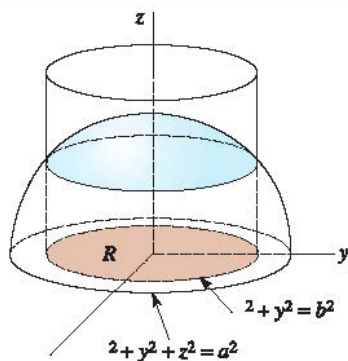


FIGURE 9.13.2 Portion of a sphere in Example 1

□ Differential of Surface Area

The function

$$dS = \sqrt{1 + [f_x(x, y)]^2 + [f_y(x, y)]^2} dA \quad (3)$$

is called the **differential of the surface area**. We will use this function in the discussion that follows.

□ Surface Integral

As we have seen, a double integral

$$\iint_R f(x, y) dA$$

is a generalization of the definite integral $\int_a^b f(x) dx$. The surface area integral (2) is a generalization of the arc length integral (1). We are now going to consider a generalization of the line integral $\int_C G(x, y) ds$. This generalization is called a **surface integral**.

1. Let $w = G(x, y, z)$ be defined in a region of 3-space that contains a surface S , which is the graph of a function $z = f(x, y)$. Let the projection R of the surface onto the xy -plane be either a Type I or a Type II region.
2. Divide the surface S into n patches S_k with areas ΔS_k that correspond to a partition P of R into n rectangles R_k with areas ΔA_k .
3. Let $\|P\|$ be the **norm** of the partition or the length of the longest diagonal of the R_k .
4. Choose a sample point (x_k^*, y_k^*, z_k^*) on each patch S_k of surface area. See **FIGURE 9.13.3**.
5. Form the sum $\sum_{k=1}^n G(x_k^*, y_k^*, z_k^*) \Delta S_k$.

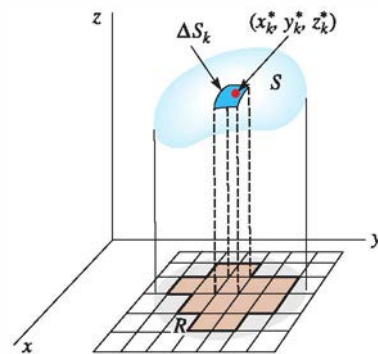


FIGURE 9.13.3 Sample point on k th patch

Definition 9.13.2 Surface Integral

Let G be a function of three variables defined over a region of 3-space containing the surface S . Then the **surface integral of G over S** is given by

$$\iint_S G(x, y, z) dS = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n G(x_k^*, y_k^*, z_k^*) \Delta S_k. \quad (4)$$

□ **Method of Evaluation** If G, f, f_x , and f_y are continuous throughout a region containing S , we can evaluate (4) by means of a double integral. From (3) the left side of (4) becomes

$$\iint_S G(x, y, z) dS = \iint_R G(x, y, f(x, y)) \sqrt{1 + [f_x(x, y)]^2 + [f_y(x, y)]^2} dA. \quad (5)$$

Note that when $G = 1$, (5) reduces to formula (2) for surface area.

□ **Projection of S Into Other Planes** If $y = g(x, z)$ is the equation of a surface S that projects onto a region R of the xz -plane, then

$$\iint_S G(x, y, z) dS = \iint_R G(x, g(x, z), z) \sqrt{1 + [g_x(x, z)]^2 + [g_z(x, z)]^2} dA. \quad (6)$$

Similarly, if $x = h(y, z)$ is the equation of a surface that projects onto the yz -plane, then the analogue of (5) is

$$\iint_S G(x, y, z) dS = \iint_R G(h(y, z), y, z) \sqrt{1 + [h_y(y, z)]^2 + [h_z(y, z)]^2} dA. \quad (7)$$

□ **Mass of a Surface** Suppose $\rho(x, y, z)$ represents the density of a surface at any point, or mass per unit surface area; then the **mass m** of the surface is

$$m = \iint_S \rho(x, y, z) dS. \quad (8)$$

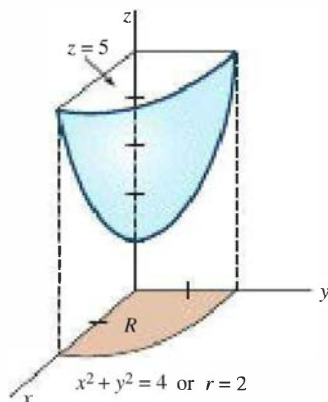


FIGURE 9.13.4 Surface in Example 2

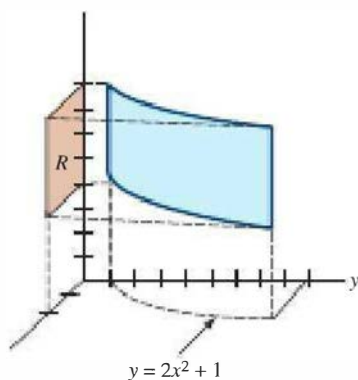
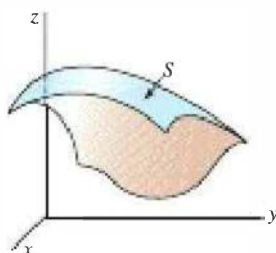


FIGURE 9.13.5 Surface in Example 3



(a) Two-sided surface



(b) One-sided surface

FIGURE 9.13.6 Oriented surface in (a); non-oriented surface in (b)

EXAMPLE 2 Mass of a Surface

Find the mass of the surface of the paraboloid $z = 1 + x^2 + y^2$ in the first octant for $1 \leq z \leq 5$ if the density at a point P on the surface is directly proportional to its distance from the xy -plane.

SOLUTION The surface in question and its projection onto the xy -plane are shown in FIGURE 9.13.4. Now, since $\rho(x, y, z) = kz$ and $z = 1 + x^2 + y^2$, (8) and (5) give

$$m = \iint_S kz \, dS = k \iint_R (1 + x^2 + y^2) \sqrt{1 + 4x^2 + 4y^2} \, dA.$$

By changing to polar coordinates, we obtain

$$\begin{aligned} m &= k \int_0^{\pi/2} \int_0^2 (1 + r^2) \sqrt{1 + 4r^2} r \, dr \, d\theta \\ &= k \int_0^{\pi/2} \int_0^2 [r(1 + 4r^2)^{1/2} + r^3(1 + 4r^2)^{1/2}] \, dr \, d\theta \quad \leftarrow \text{integration by parts} \\ &= k \int_0^{\pi/2} \left[\frac{1}{12} (1 + 4r^2)^{3/2} + \frac{1}{12} r^2(1 + 4r^2)^{3/2} - \frac{1}{120} (1 + 4r^2)^{5/2} \right]_0^2 d\theta \\ &= \frac{k\pi}{2} \left[\frac{5(17)^{3/2}}{12} - \frac{17^{5/2}}{120} - \frac{3}{40} \right] \approx 30.16 k. \end{aligned}$$

EXAMPLE 3 Evaluating a Surface Integral

Evaluate $\iint_S xz^2 \, dS$, where S is that portion of the cylinder $y = 2x^2 + 1$ in the first octant bounded by $x = 0$, $x = 2$, $z = 4$, and $z = 8$.

SOLUTION We shall use (6) with $g(x, z) = 2x^2 + 1$ and R the rectangular region in the xz -plane shown in FIGURE 9.13.5. Since $g_x(x, z) = 4x$ and $g_z(x, z) = 0$, it follows that

$$\begin{aligned} \iint_S xz^2 \, dS &= \int_0^2 \int_4^8 xz^2 \sqrt{1 + 16x^2} \, dz \, dx \\ &= \int_0^2 \left[\frac{z^3}{3} x \sqrt{1 + 16x^2} \right]_4^8 dx = \frac{448}{3} \int_0^2 x(1 + 16x^2)^{1/2} dx \\ &= \frac{28}{9} (1 + 16x^2)^{3/2} \Big|_0^2 = \frac{28}{9} [65^{3/2} - 1] \approx 1627.3. \end{aligned}$$

Orientable Surfaces In Example 5, we are going to evaluate a surface integral of a vector field. In order to do this we need to examine the concept of an **orientable surface**. Roughly, an orientable surface S , such as that given in FIGURE 9.13.6(a), has two sides that could be painted different colors. The Möbius strip* shown in Figure 9.13.6(b) is not an orientable surface and is one-sided. A person who starts to paint the surface of a Möbius strip at a point will paint the entire surface and return to the starting point.

Specifically, we say a smooth surface S is **orientable** or is an **oriented surface** if there exists a continuous unit normal vector function $\mathbf{n}$ defined at each point (x, y, z) on the surface. The vector field $\mathbf{n}(x, y, z)$ is called the **orientation** of S . But since a unit normal to the surface S at (x, y, z) can be either $\mathbf{n}(x, y, z)$ or $-\mathbf{n}(x, y, z)$, an orientable surface has two orientations. See FIGURE 9.13.7(a)–(c). The Möbius strip shown again in Figure 9.13.7(d) is not an oriented surface, since if a unit normal $\mathbf{n}$ starts at P on the surface and moves *once* around the strip on the curve C , it ends up on the “opposite side” of the strip at P and so points in the opposite direction.

*To construct a Möbius strip cut out a long strip of paper, give one end a half-twist, and then attach the ends with tape.

A surface S defined by $z = f(x, y)$ has an **upward orientation** (Figure 9.13.7(b)) when the unit normals are directed upward—that is, have positive $\mathbf{k}$ components, and it has a **downward orientation** (Figure 9.13.7(c)) when the unit normals are directed downward—that is, have negative $\mathbf{k}$ components.

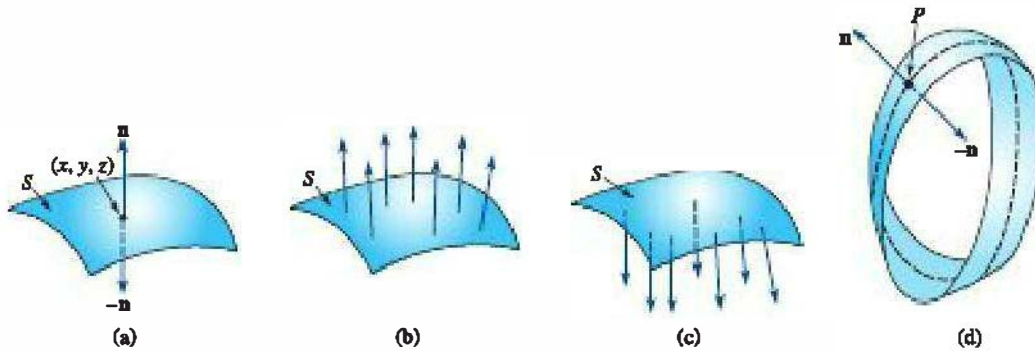


FIGURE 9.13.7 Upward orientation in (b); downward orientation in (c)

If a smooth surface S is defined by $g(x, y, z) = 0$, then recall that a unit normal is

$$\mathbf{n} = \frac{1}{\|\nabla g\|} \nabla g, \quad (9)$$

where $\nabla g = \frac{\partial g}{\partial x} \mathbf{i} + \frac{\partial g}{\partial y} \mathbf{j} + \frac{\partial g}{\partial z} \mathbf{k}$ is the gradient of g . If S is defined by $z = f(x, y)$, then we can use $g(x, y, z) = z - f(x, y) = 0$ or $g(x, y, z) = f(x, y) - z = 0$ depending on the orientation of S .

As we shall see in the next example, the two orientations of an orientable closed surface are outward and inward.

EXAMPLE 4 Orientations of a Surface

Consider the sphere of radius $a > 0$: $x^2 + y^2 + z^2 = a^2$. If we define $g(x, y, z) = x^2 + y^2 + z^2 - a^2$, then

$$\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} \quad \text{and} \quad \|\nabla g\| = \sqrt{4x^2 + 4y^2 + 4z^2} = 2a.$$

Then the two orientations of the surface are

$$\mathbf{n} = \frac{x}{a}\mathbf{i} + \frac{y}{a}\mathbf{j} + \frac{z}{a}\mathbf{k} \quad \text{and} \quad \mathbf{n}_1 = -\mathbf{n} = -\frac{x}{a}\mathbf{i} - \frac{y}{a}\mathbf{j} - \frac{z}{a}\mathbf{k}.$$

The vector field $\mathbf{n}$ defines an outward orientation, whereas $\mathbf{n}_1 = -\mathbf{n}$ defines an inward orientation. See **FIGURE 9.13.8**.

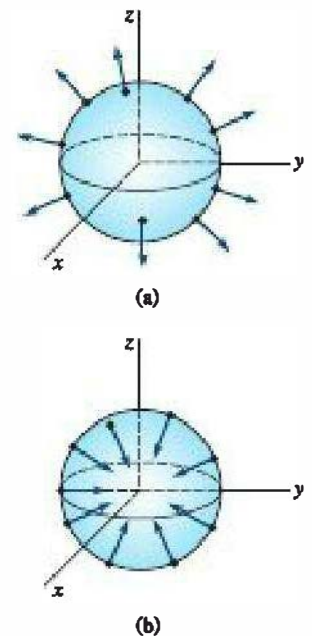


FIGURE 9.13.8 Sphere in Example 4

Integrals of Vector Fields If $\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ is the velocity field of a fluid, then, as we saw in Figure 9.7.3, the volume of the fluid flowing through an element of surface area ΔS per unit time is approximated by

$$(\text{height})(\text{area of base}) = (\text{comp}_{\mathbf{n}} \mathbf{F}) \Delta S = (\mathbf{F} \cdot \mathbf{n}) \Delta S,$$

where $\mathbf{n}$ is a unit normal to the surface. See **FIGURE 9.13.9**. The total volume of a fluid passing through S per unit time is called the **flux** of $\mathbf{F}$ through S and is given by

$$\text{flux} = \iint_S (\mathbf{F} \cdot \mathbf{n}) dS. \quad (10)$$

In the case of a closed surface S , if $\mathbf{n}$ is the outer (inner) normal, then (10) gives the volume of fluid flowing out (in) through S per unit time.

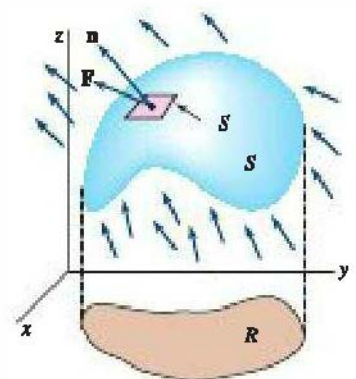


FIGURE 9.13.9 Surface S in (10)

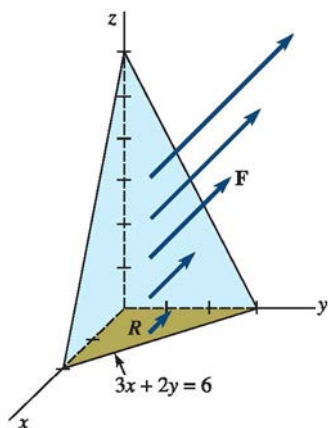


FIGURE 9.13.10 Surface in Example 5

EXAMPLE 5 Flux Through a Surface

Let $\mathbf{F}(x, y, z) = z\mathbf{j} + z\mathbf{k}$ represent the flow of a liquid. Find the flux of $\mathbf{F}$ through the surface S given by that portion of the plane $z = 6 - 3x - 2y$ in the first octant oriented upward.

SOLUTION The vector field and the surface are illustrated in FIGURE 9.13.10. By defining the plane by $g(x, y, z) = 3x + 2y + z - 6 = 0$, we see that a unit normal with a positive $\mathbf{k}$ component is

$$\mathbf{n} = \frac{\nabla g}{\|\nabla g\|} = \frac{3}{\sqrt{14}}\mathbf{i} + \frac{2}{\sqrt{14}}\mathbf{j} + \frac{1}{\sqrt{14}}\mathbf{k}.$$

Hence,

$$\text{flux} = \iint_S (\mathbf{F} \cdot \mathbf{n}) \, dS = \frac{1}{\sqrt{14}} \iint_S 3z \, dS.$$

With R the projection of the surface onto the xy -plane, we find from (10) that

$$\begin{aligned} \text{flux} &= \frac{1}{\sqrt{14}} \iint_R 3(6 - 3x - 2y)(\sqrt{14} \, dA) \\ &= 3 \int_0^2 \int_0^{3-3x/2} (6 - 3x - 2y) \, dy \, dx = 18. \end{aligned}$$

Depending on the nature of the vector field, the integral in (10) can represent other kinds of flux. For example, (10) could also give electric flux, magnetic flux, flux of heat, and so on.

Remarks

If the surface S is piecewise defined, we express a surface integral over S as the sum of the surface integrals over the various pieces of the surface. For example, suppose S is the orientable piecewise-smooth closed surface bounded by the paraboloid $z = x^2 + y^2$ (S_1) and the plane $z = 1$ (S_2). Then the flux of a vector field $\mathbf{F}$ out of the surface S is

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_{S_1} \mathbf{F} \cdot \mathbf{n} \, dS + \iint_{S_2} \mathbf{F} \cdot \mathbf{n} \, dS,$$

where we take S_1 oriented upward and S_2 oriented downward. See FIGURE 9.13.11 and Problem 35 in Exercises 9.13.

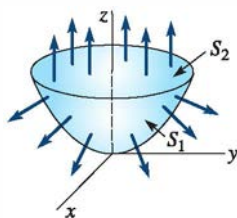


FIGURE 9.13.11 Piecewise-defined surface

9.13 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

- Find the surface area of that portion of the plane $2x + 3y + 4z = 12$ that is bounded by the coordinate planes in the first octant.
- Find the surface area of that portion of the plane $2x + 3y + 4z = 12$ that is above the region in the first quadrant bounded by the graph $r = \sin 2\theta$.
- Find the surface area of that portion of the cylinder $x^2 + z^2 = 16$ that is above the region in the first quadrant bounded on the graphs of $x = 0$, $x = 2$, $y = 0$, $y = 5$.
- Find the surface area of that portion of the paraboloid $z = x^2 + y^2$ that is below the plane $z = 2$.
- Find the surface area of that portion of the paraboloid $z = 4 - x^2 - y^2$ that is above the xy -plane.
- Find the surface area of those portions of the sphere $x^2 + y^2 + z^2 = 2$ that are within the cone $z^2 = x^2 + y^2$.
- Find the surface area of the portion of the sphere $x^2 + y^2 + z^2 = 25$ that is above the region in the first quadrant bounded by the graphs of $x = 0$, $y = 0$, $4x^2 + y^2 = 25$. [Hint: Integrate first with respect to x .]
- Find the surface area of that portion of the graph of $z = x^2 - y^2$ that is in the first octant within the cylinder $x^2 + y^2 = 4$.
- Find the surface area of the portions of the sphere $x^2 + y^2 + z^2 = a^2$ that are within the cylinder $x^2 + y^2 = ay$.
- Find the surface area of the portions of the cone $z^2 = \frac{1}{4}(x^2 + y^2)$ that are within the cylinder $(x - 1)^2 + y^2 = 1$.
- Find the surface area of the portions of the cylinder $y^2 + z^2 = a^2$ that are within the cylinder $x^2 + y^2 = a^2$. [Hint: See Figure 9.10.12.]
- Use the result given in Example 1 to prove that the surface area of a sphere of radius a is $4\pi a^2$. [Hint: Consider a limit as $b \rightarrow a$.]
- Find the surface area of that portion of the sphere $x^2 + y^2 + z^2 = a^2$ that is bounded between $y = c_1$ and $y = c_2$, $0 < c_1 < c_2 < a$. [Hint: Use polar coordinates in the xz -plane.]
- Show that the area found in Problem 13 is the same as the surface area of the cylinder $x^2 + z^2 = a^2$ between $y = c_1$ and $y = c_2$.